

Utkarsh Sonawane

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EDUCATION

The Pennsylvania State University

Bachelor of Science in Computer Science

Harrisburg, PA

Expected December 2028

- **GPA:** 3.53/4.0 | **Honors:** Dean's List, Capital Honors Program
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Calculus III, Linear Algebra

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, Swift, SQL
Backend & Systems: FastAPI, Node.js, Express, Flask, REST APIs, Docker, Linux, Git
Data & Databases: PostgreSQL, Supabase, pgvector, Vector Embeddings, RAG, NumPy, Pandas
Mobile & Client: SwiftUI, HealthKit, React, Next.js, Tailwind CSS

PROJECTS

Nytr — Evidence-Driven Decision System | *Python, FastAPI, PostgreSQL, SwiftUI, HealthKit* 2026

- Engineered an evidence-driven decision system integrating Penn State dining menus, Apple HealthKit, and Hevy through a FastAPI modular backend and native SwiftUI companion client.
- Designed immutable/versioned pipelines for nutrition history, 28-day Theil–Sen body-mass trends, target policies, and workout revisions with PostgreSQL RLS, idempotent sync via durable anchors, and deterministic Decimal analytics.
- Enforced strict dietary eligibility gates, deterministic Decimal nutrition calculations, and tombstone deletion semantics; validated production correctness with 1,000+ automated tests and physical-device verification.

Clage — From-Scratch NEAT Engine & Artificial Life | *Python, Algorithms from Scratch* 2025 – 2026

- Implemented complete NeuroEvolution of Augmenting Topologies from first principles in pure Python (genome encoding, global innovation ledger, topological crossover, mutation, and speciation) with zero external ML libraries.
- Validated the evolutionary engine on OR, AND, XOR, and sine benchmarks across fixed seeds before deploying in a 2D artificial-life simulation with dynamic energy budgets and reproduction, backed by 188 passing tests.
- Investigated behavioral diversity metrics across populations; identified and eliminated a local food density confound from cross-condition comparative analysis.

Devvy — Developer Infrastructure for Discord IPC | *TypeScript, Node.js, macOS launchd* 2026

- Built a zero-dependency background daemon speaking Discord's local IPC protocol directly (Unix socket discovery, binary opcode framing, backoff reconnect) to coordinate Rich Presence across 3 developer environments.
- Designed priority arbitration (OpenCode → Command Code → VS Code) with TTL-based heartbeat expiry; validated with a 22-script test suite covering race conditions, reconnect resilience, and IPC socket lifecycle.

ScholarAI — Government Scheme Discovery Platform | *Python, PostgreSQL, pgvector, FastAPI* 2026

- Data engineer on a 5-person team; finalist among 6,081+ global participants in the USAII Global AI Hackathon 2026.
- Built data pipeline parsing 1,000+ scraped schemes into structured relational models; generated high-dimensional semantic embeddings and loaded them via fault-tolerant batched inserts into PostgreSQL with pgvector for hybrid retrieval.

LEADERSHIP & ACTIVITIES

CS Team Lead & Technical Team Member, IEEE

IEEE Student Branch, Penn State Harrisburg

Harrisburg, PA

Sept 2025 – Present

- Conducted hands-on technical workshop for 20+ students on documentation, Markdown, and developer toolchains.
- Lead technical initiatives and collaborative coding sessions across Computer Science and AI student branches.

Founder & Editor-in-Chief

Glimpse — Student Publication

Dubai, UAE

Jun 2024 – Feb 2025

- Scaled subscriber readership from 0 to 250+ readers; authored and edited 30+ long-form articles on technology and science.

CERTIFICATIONS & HONORS

Honors: Finalist, USAII Global AI Hackathon (Top teams among 6,081+ participants) | Dean's List, Penn State

Certifications: Practical Deep Learning for Coders (*fast.ai*) | Machine Learning Specialization (*DeepLearning.ai*) | Ethics of AI (*University of Helsinki*)